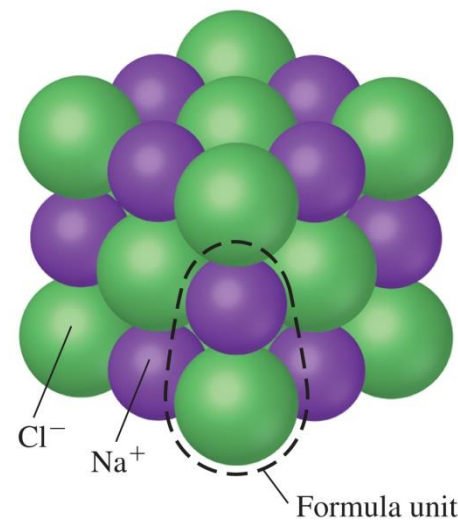


GENERAL CHEMISTRY I



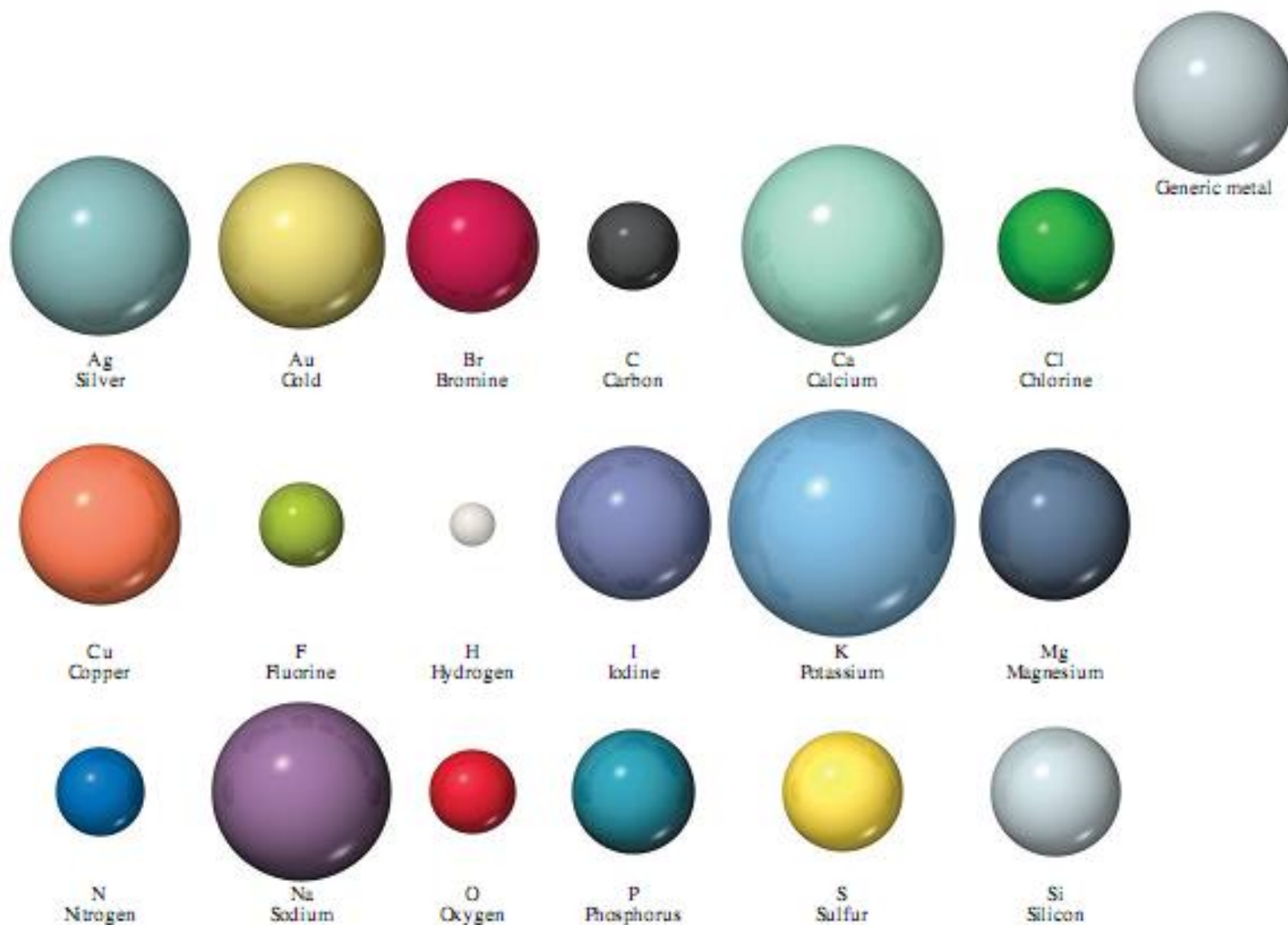
Chapter 3

Chemical Reactions and Reaction stoichiometry

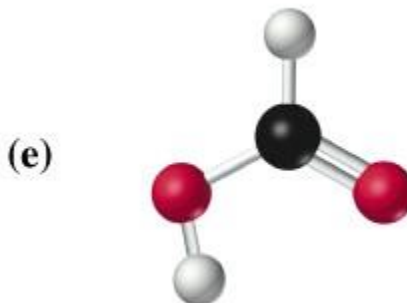
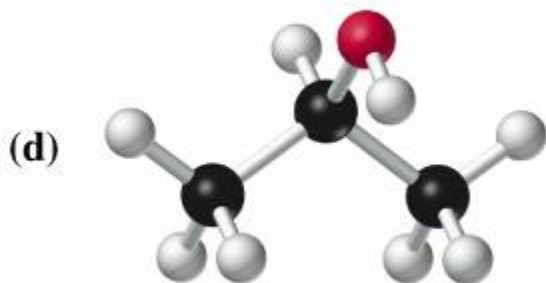
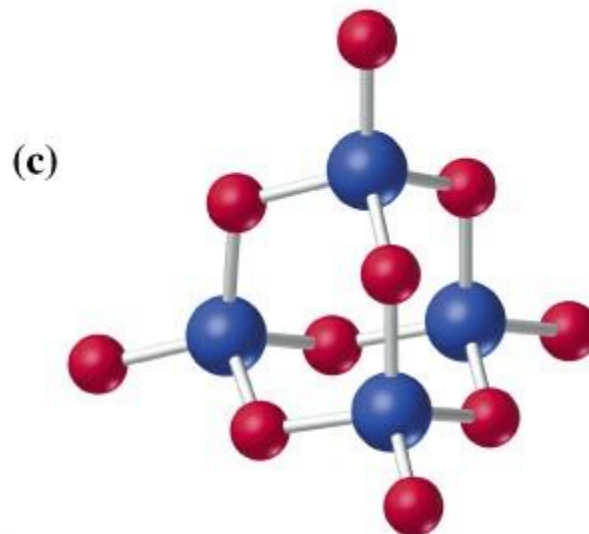
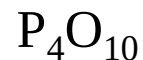
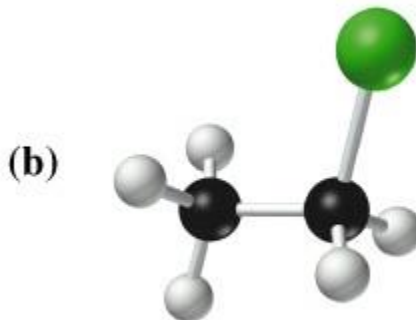
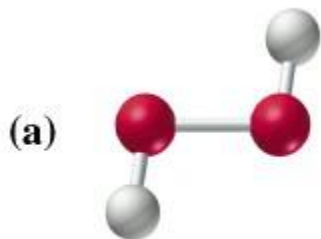
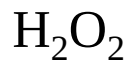
Contents

- 3-1 Chemical Equations
- 3-2 Simple Patterns of Chemical Reactivity
- 3-3 Formula Weights
- 3-4 Avogadro's Number and the Mole
- 3-5 Empirical Formulas from Analyses
- 3-6 Quantitative Information from Balanced Equations
- 3-7 Limiting Reactants

Color Chart for Common Elements



Some Organic and Inorganic Molecules



Copper Penny Reacting with Nitric Acid in a Test Tube



<https://www.youtube.com/watch?v=uOa2v0Y0leU>

As **reactants** are converted to **products** we observe:

- Color change
- Precipitate formation
- Gas evolution
- Heat absorption or evolution

Chemical evidence may be necessary.

Stoichiometry



Antoine Lavoisier
(1743 – 1794)

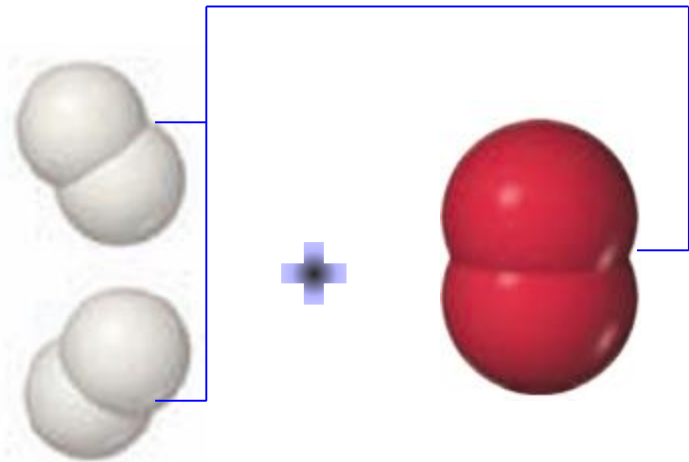
- Stoichiometry is built on an understanding of
 - atomic masses,
 - chemical formulas,
 - the law of conservation of mass.

*“We may lay it down as an incontestable axiom that, in all the operations of art and nature, **nothing is created; an equal quantity of matter exists both before and after the experiment.** Upon this principle, the whole art of performing chemical experiments depends.”*

3-1 Chemical Equations

Hydrogen + Oxygen  Water

These covalent bonds break

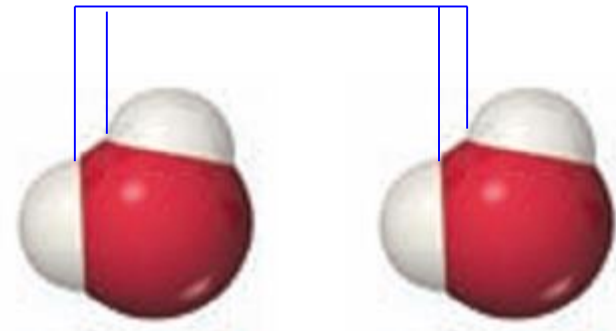


4 H

2O

same numbers and kinds of atoms
on both side of the arrow

New covalent bonds form



4H

2O

the coefficient

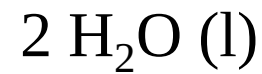


physical state

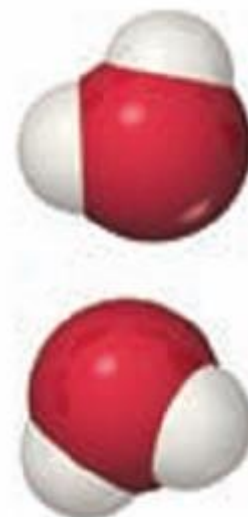
*special conditions
(like heat or light)*



the coefficient



physical state

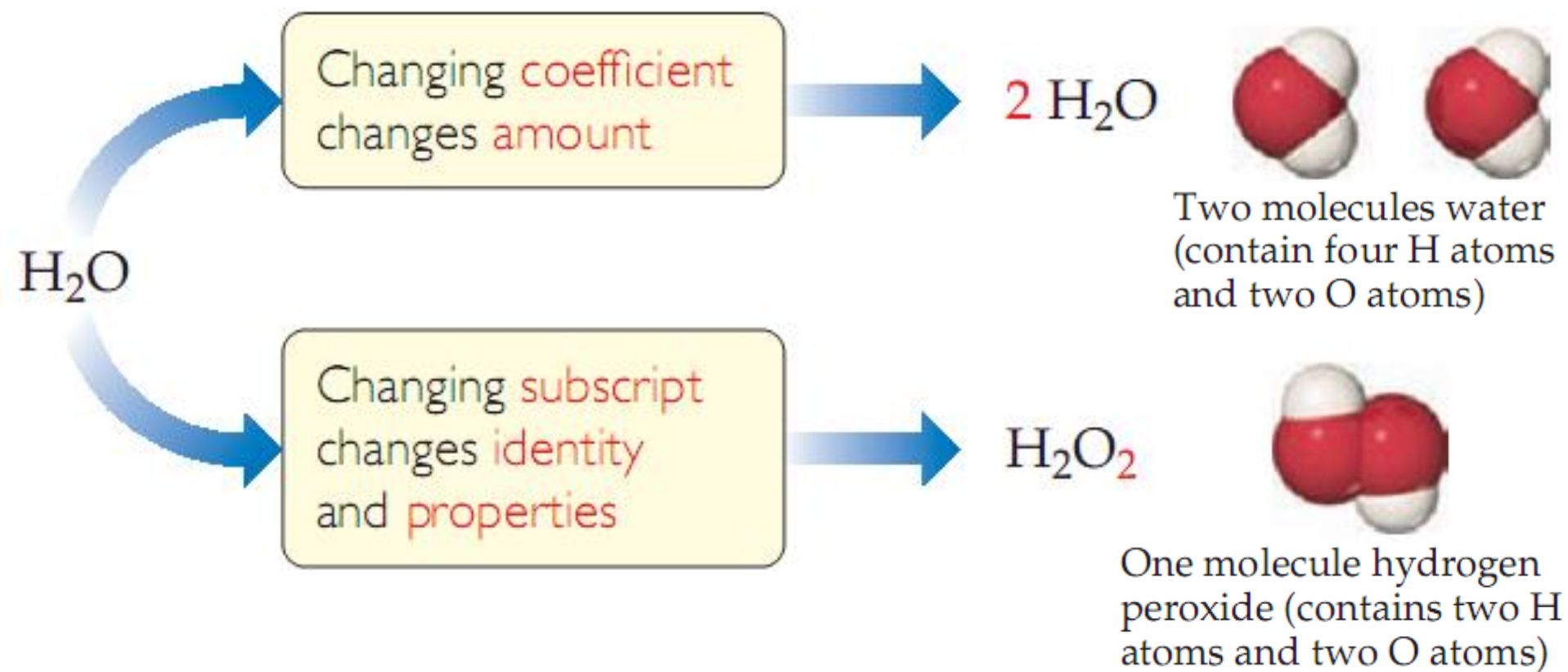


This arrow means
“goes to”,
“yields”, or
“produces”

reactants

products

Two molecules of hydrogen **react with** one molecule of oxygen **produces** two molecules of water

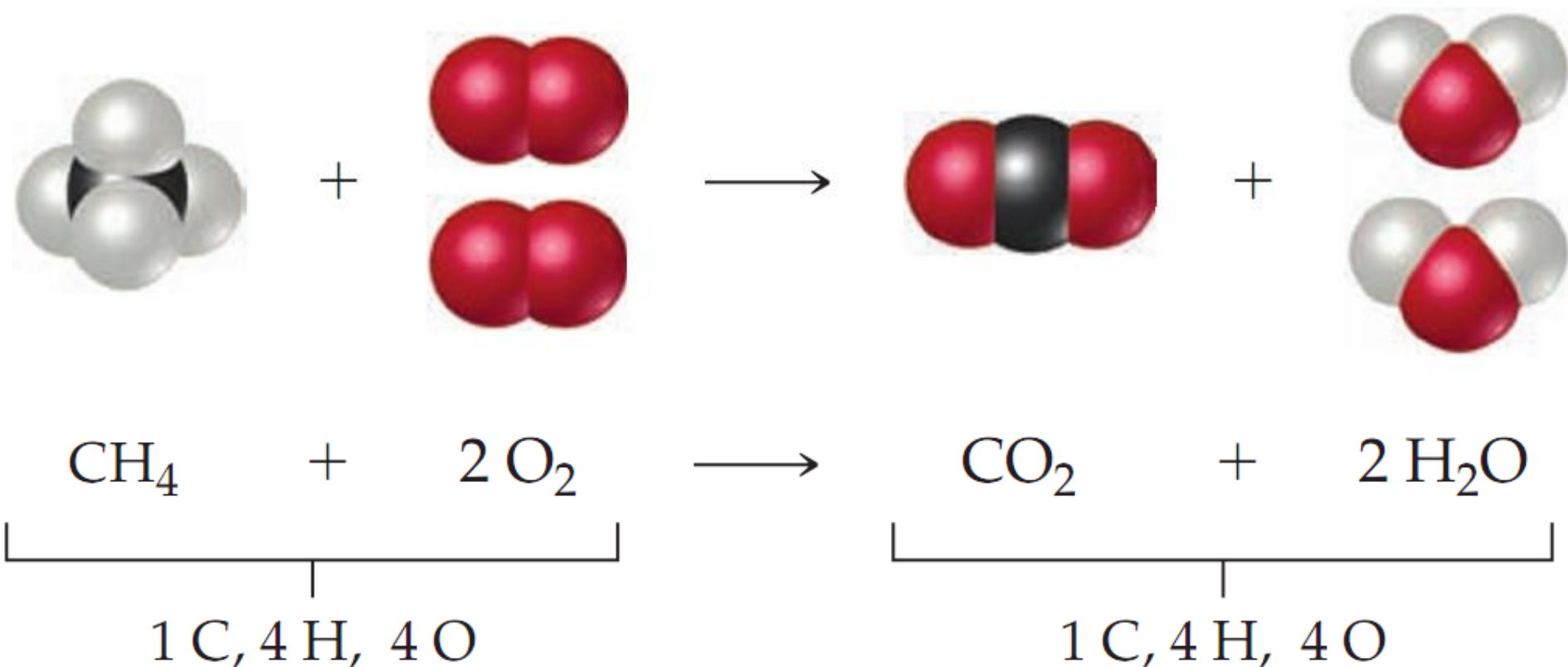


Balancing Equations

When balancing an equation always make sure the **type** and **number** of atoms on both sides of the arrow are equal.

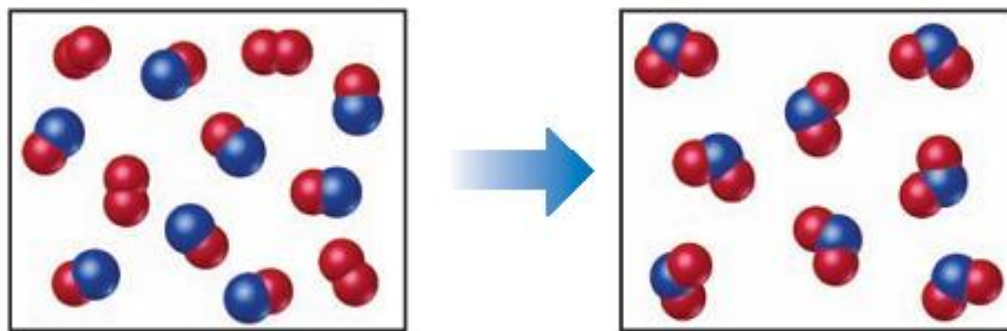
How many C, H, and O atoms are present on the reactant side?

Are the same number of each type of atom present on the product side?



EXAMPLE

The following diagram represents a chemical reaction in which the red spheres are oxygen atoms and the blue spheres are nitrogen atoms. The reactants and products are in gaseous state.

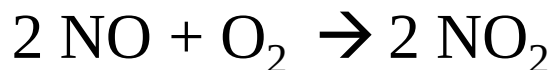


(a) Write the chemical formulas for the reactants and products.

Reactants: NO and O₂

Product: NO₂

(b) Write a balanced equation for the reaction.



(c) Is the diagram consistent with the law of conservation of mass?

Yes, it is. There are 16 O and 8 N on both sides

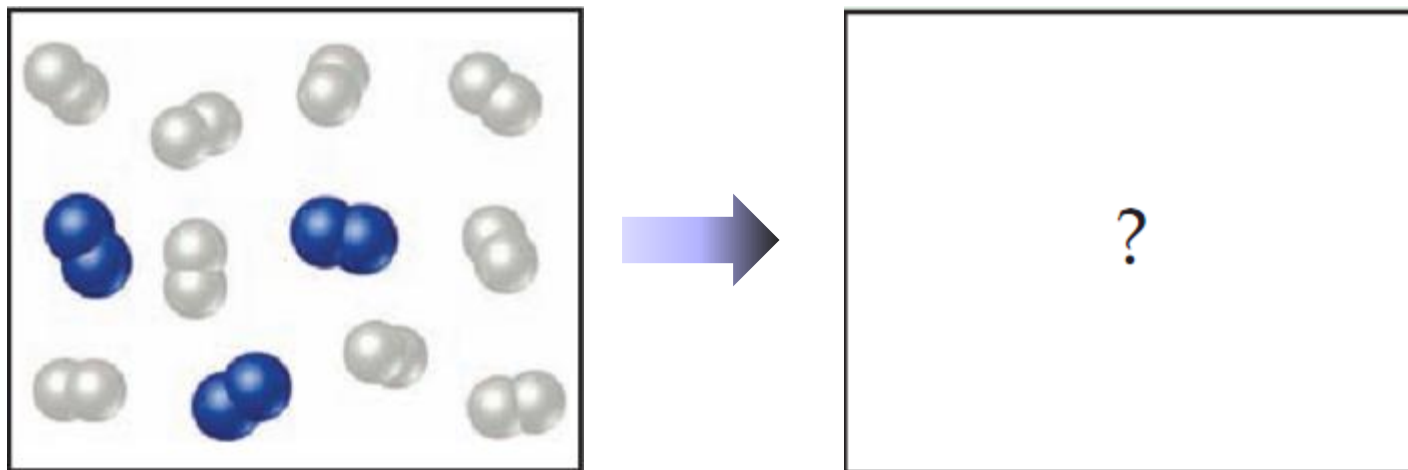
PRACTICE EXERCISE 1

In the following diagram, the white spheres represent hydrogen atoms and the blue spheres represent nitrogen atoms.

The two reactants combine to form a single product, ammonia, NH_3 , which is not shown.

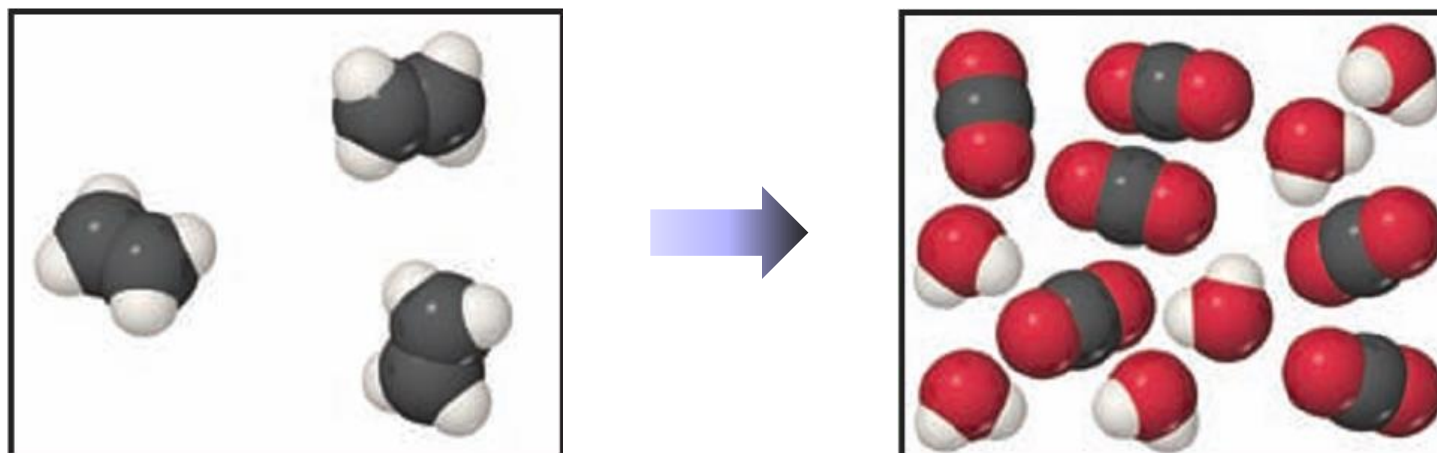
Write a balanced chemical equation for the reaction. Based on the equation and the contents of the left (reactants) box, find how many NH_3 molecules should be shown in the right (products) box.

- (a) 2
- (b) 3
- (c) 4
- (d) 6
- (e) 9

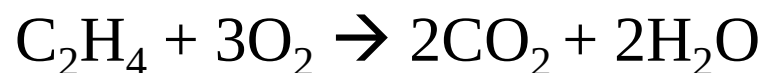


PRACTICE EXERCISE 2

In the following diagram, the white spheres represent hydrogen atoms, the black spheres carbon atoms, and the red spheres oxygen atoms.



(a) Write a balanced chemical equation for the reaction.



(b) Determine the number of O_2 molecules that should be shown in the left (reactants) box.

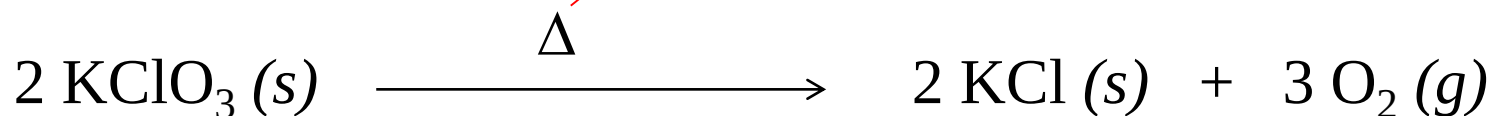


Indicating the States of Reactants and Products

This indicates a special condition



This indicates the continuous addition of heat



- (g) gases
- (l) liquids
- (s) solids
- (aq) dissolved in aqueous (water) solution

EXAMPLE

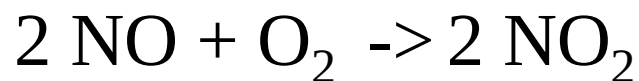


Balance the equation

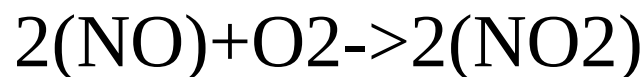


For the online quiz where typing subscription might be impossible, especially **on a smartphone**, I'm okay with putting the molecule in parenthesis.

For example, instead of filling in the blank with



You can use

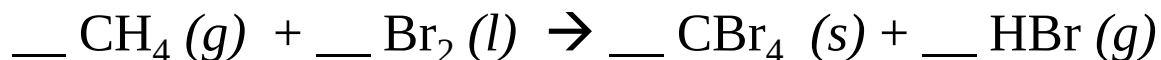


For the online quiz, do not add space to the equations!

It's time for a quiz!

PRACTICE EXERCISE 1

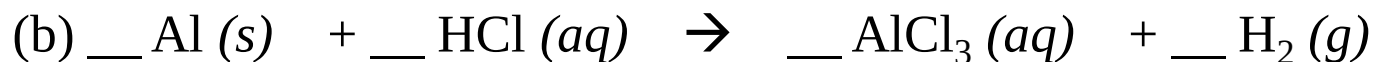
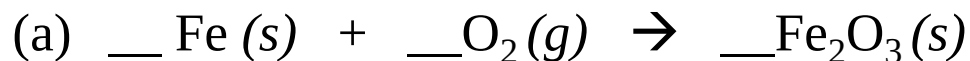
The unbalanced equation for the reaction between methane and bromine is



Once this equation is balanced what is the value of the coefficient in front of bromine Br₂?

PRACTICE EXERCISE 2

Balance these equations by providing the missing coefficients:



3-2 Simple Patterns of Chemical Reactivity



1. Synthesis (Combination)
2. Decomposition
3. Single replacement
4. Double replacement
5. Combustion

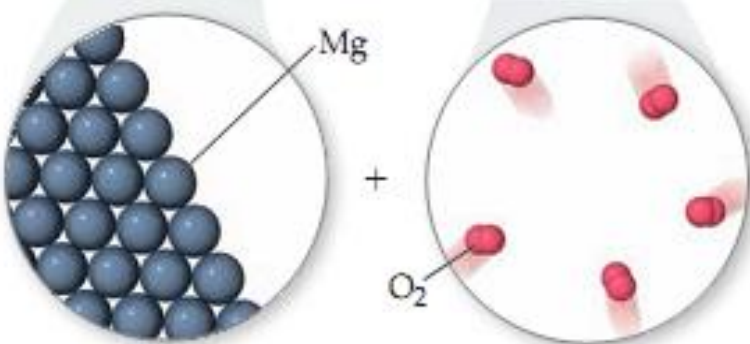
Combination Reactions

Two or more reactants combine to form a single product.



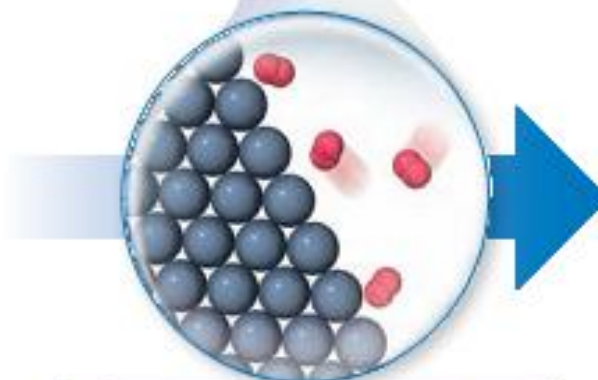
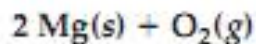
It was such a great combination of brass and bamboo!

Combustion of magnesium metal in air, a combination reaction

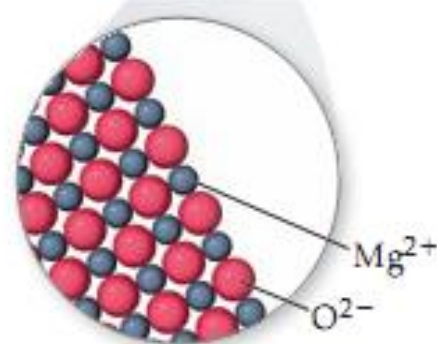


The ribbon of magnesium metal is surrounded by oxygen gas in the air.

Reactants

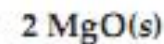


An intense flame is produced as the Mg atoms react with O₂.



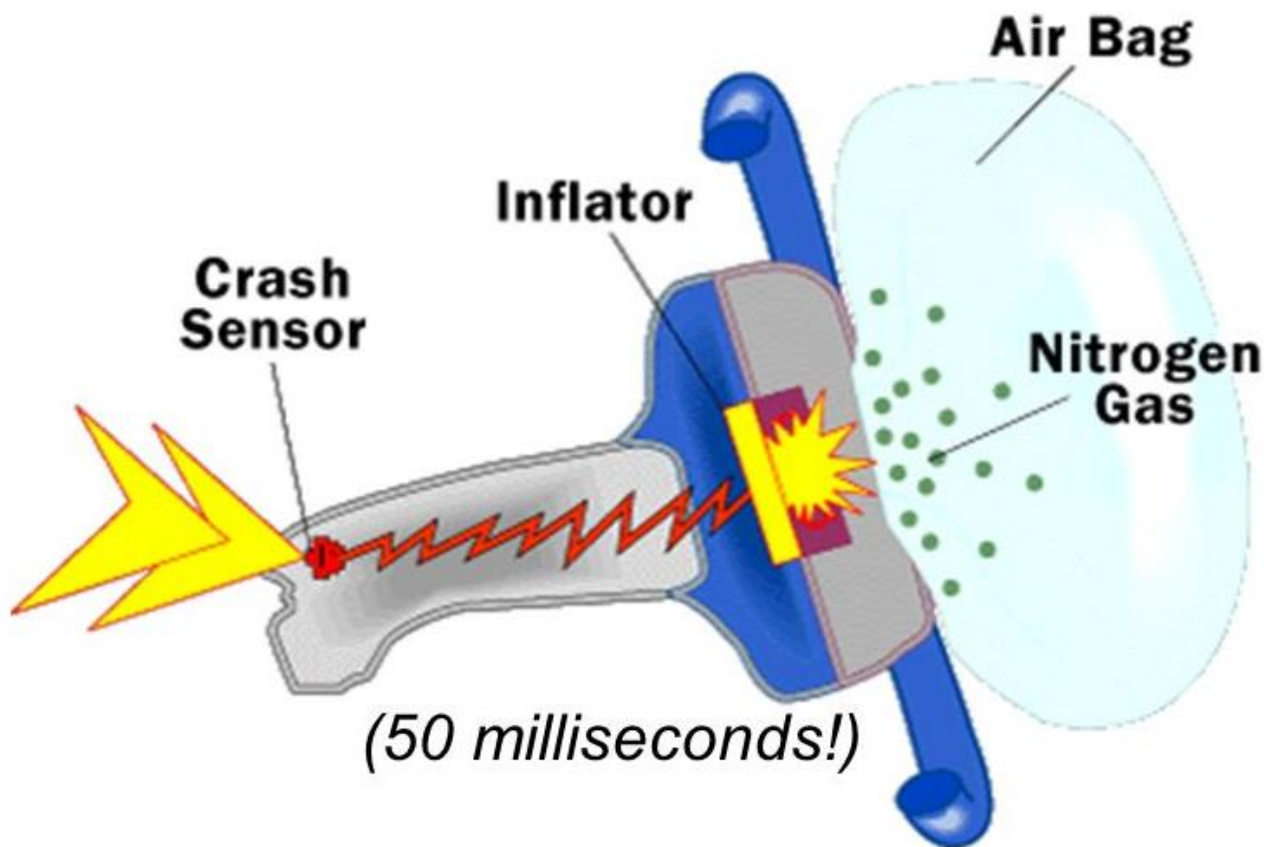
The reaction forms MgO, a white, ionic solid.

Products

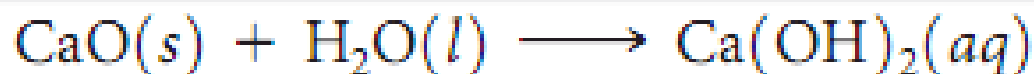


Decomposition Reactions

A single reactant undergoes a reaction to produce two or more other substances



Combination Reactions



Decomposition Reactions



EXAMPLE

Write a balanced equation for

(a) the combination reaction between lithium metal and fluorine gas

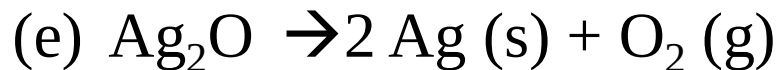
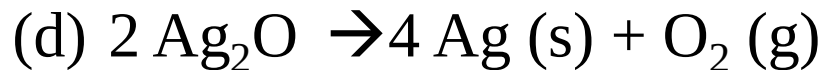
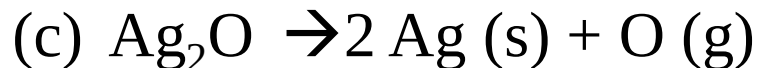
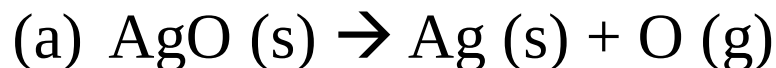


(b) the decomposition reaction that occurs when solid barium carbonate is heated (two products form, a solid and a gas).



PRACTICE 1

Which of the following reactions is the balanced equation that represents the decomposition reaction that occurs when silver (I) oxide is heated?



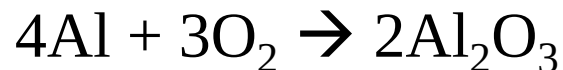
PRACTICE 2

Write a balanced equation for

(a) solid mercury(II) sulfide decomposing into its component elements when heated



(b) aluminum metal combining with oxygen in the air.



Combustion Reactions

Combustion reactions are rapid reactions that produce a flame. Most combustion reactions we observe involve O_2 from air as a reactant.



pure methanol vs pure ethanol
in combustion

Why?

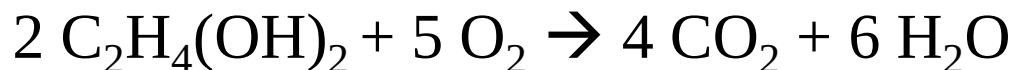
PRACTICE

Write the balanced equation for the reaction that occurs when

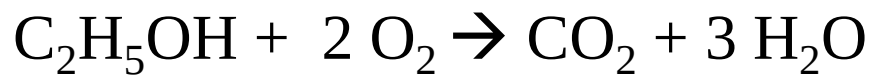
(a) methanol, CH_3OH (*l*), is burned in air.



(b) ethylene glycol, $\text{C}_2\text{H}_4(\text{OH})_2$ (*l*), burns in air.



(c) ethanol, $\text{C}_2\text{H}_5\text{OH}$ (*l*), burns in air.



Behind the fireworks

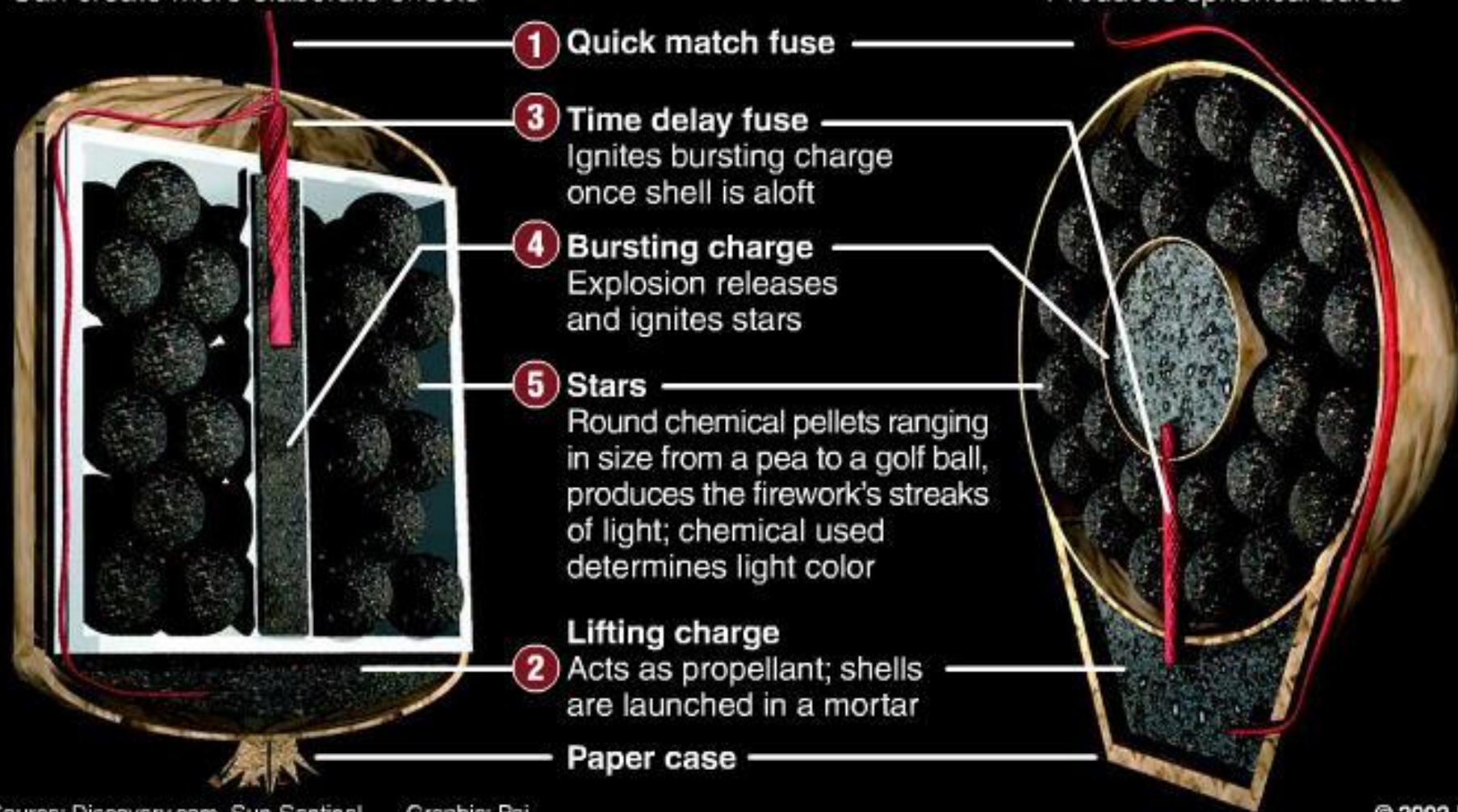
Two main types of aerial shells used for fireworks shows:

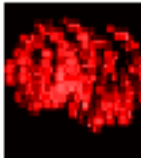
Italian-style shell

Can create more elaborate effects

Oriental-style shell

Produces spherical bursts



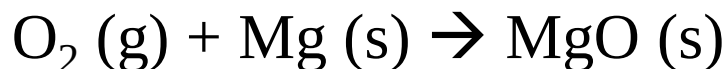
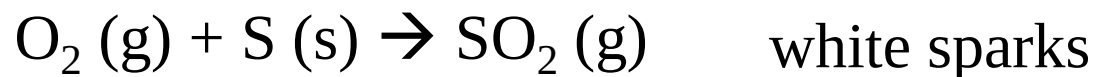
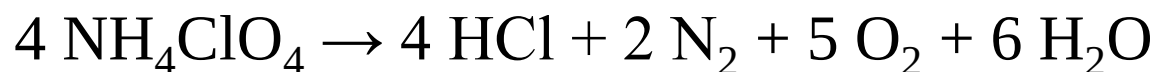
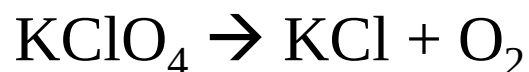
	Color	Compound	Wavelength (nm)
	red	strontium salts, lithium salts lithium carbonate, Li_2CO_3 = deep red strontium carbonate, SrCO_3 = bright red	650
	orange	calcium salts calcium chloride, CaCl_2	670
	yellow	sodium salts sodium chloride, NaCl	610-620
	green	barium compounds + chlorine producer barium chloride, BaCl_2	590
	blue	copper compounds + chlorine producer copper(I) chloride, CuCl	500-535
	purple	mixture of strontium (red) and copper (blue) compounds	420-460
	silver	burning metallic aluminum, titanium, or magnesium	all

Chemical Reactions in the Black Powder

- Black powder (bursting and lifting charges): 75% potassium nitrate, 15% charcoal, and 10% sulfur
 - Decomposition reaction of the oxidizers to produce oxygen
$$4 \text{KNO}_3 (\text{s}) \rightarrow 2 \text{K}_2\text{O} (\text{s}) + 2 \text{N}_2 (\text{g}) + 5 \text{O}_2 (\text{g})$$
 - Combination reactions of oxygen with reducing agents to produce hot, rapidly expanding gasses
$$\text{O}_2 (\text{g}) + \text{C} (\text{s}) \rightarrow \text{CO}_2 (\text{g})$$
$$\text{O}_2 (\text{g}) + \text{S} (\text{s}) \rightarrow \text{SO}_2 (\text{g})$$

Chemical Reactions in the Stars

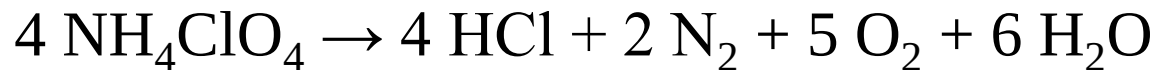
- Stars: potassium perchlorate or ammonium perchlorate, charcoal, sulfur, and metal salts
 - Decomposition reaction of the oxidizers to produce oxygen
- Combination reactions of oxygen with reducing agents to produce hot, rapidly expanding gases



Chemical Reactions in the Stars

- Stars: potassium perchlorate or ammonium perchlorate, charcoal, sulfur, and metal salts

- Decomposition reaction of the oxidizers to produce oxygen



- Combination reactions of oxygen with reducing agents to produce hot, rapidly expanding gases



3-3 Formula Weights

The **formula weight** (FW) of a substance is the **sum** of the **atomic weights** (AW) of the atoms in the chemical formula of the substance.

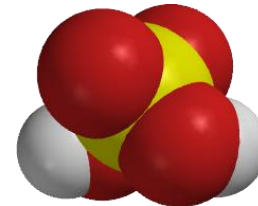
- If the chemical formula is the chemical symbol of an element,

$$FW = AW$$

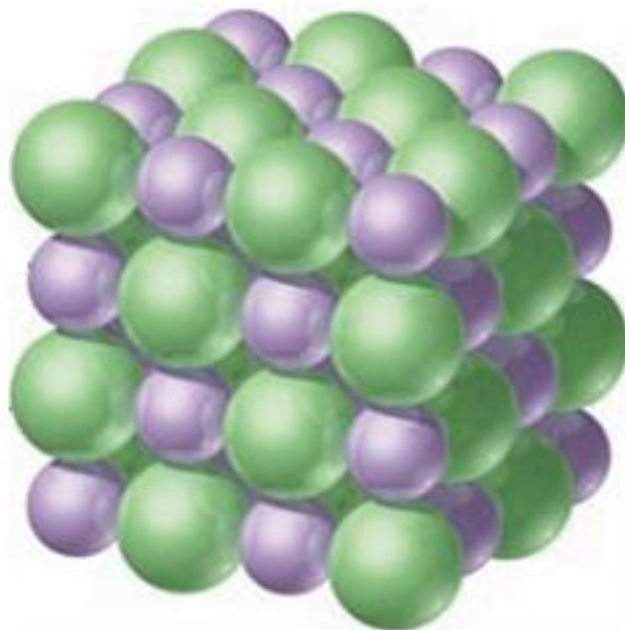
$$FW \text{ of Na} = AW \text{ of Na} = 23.0 \text{ amu}$$



- If the chemical formula is that of a molecule, the formula weight is also called the **molecular weight** (MW)



$$\begin{aligned} FW \text{ of H}_2\text{SO}_4 &= 2 (AW \text{ of H}) + (AW \text{ of S}) + 4 (AW \text{ of O}) \\ &= 2(1.0 \text{ amu}) + 32.1 \text{ amu} + 4(16.0 \text{ amu}) \\ &= 98.1 \text{ amu} \end{aligned}$$



It is inappropriate to speak of molecules of **ionic substances** the **empirical formula** is used as the formula unit.

$$\begin{aligned}\text{FW of NaCl} &= (\text{AW of Na}) + (\text{AW of Cl}) \\ &= 23.0 \text{ amu} + 35.5 \text{ amu} \\ &= 58.5 \text{ amu}\end{aligned}$$

EXAMPLE

Calculate the formula weight of

(a) sucrose, $C_{12}H_{22}O_{11}$ (table sugar);

$$12 \text{ C atoms} = 12(12.0 \text{ amu}) = 144.0 \text{ amu}$$

$$22 \text{ H atoms} = 22(1.0 \text{ amu}) = 22.0 \text{ amu}$$

$$11 \text{ O atoms} = 11(16.0 \text{ amu}) = \frac{176.0 \text{ amu}}{342.0 \text{ amu}}$$

(b) calcium nitrate, $Ca(NO_3)_2$.

$$1 \text{ Ca atom} = 1(40.1 \text{ amu}) = 40.1 \text{ amu}$$

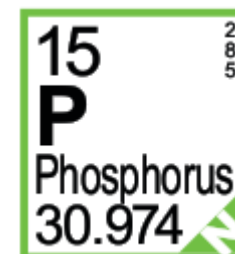
$$2 \text{ N atoms} = 2(14.0 \text{ amu}) = 28.0 \text{ amu}$$

$$6 \text{ O atoms} = 6(16.0 \text{ amu}) = \frac{96.0 \text{ amu}}{164.1 \text{ amu}}$$

PRACTICE 1

Which of the following is the correct formula weight for calcium phosphate?

- (a) 310.17 amu,
- (b) 135.11 amu,
- (c) 182.22 amu,
- (d) 278.25 amu,
- (e) 175.11 amu.



PRACTICE 2

Calculate the formula weight of



Percentage Composition from Chemical Formulas

$$\% \text{ composition of element} = \frac{\left(\begin{array}{c} \text{number of atoms} \\ \text{of element} \end{array} \right) \left(\begin{array}{c} \text{atomic weight} \\ \text{of element} \end{array} \right)}{\text{formula weight of substance}} \times 100\%$$



$$\% \text{C} = \frac{(12)(12.0 \text{ amu})}{342.0 \text{ amu}} \times 100\% = 42.1\%$$

$$\% \text{H} = \frac{(22)(1.0 \text{ amu})}{342.0 \text{ amu}} \times 100\% = 6.4\%$$

$$\% \text{O} = \frac{(11)(16.0 \text{ amu})}{342.0 \text{ amu}} \times 100\% = 51.5\%$$

PRACTICE

1. What is the percentage of nitrogen, by mass, in calcium nitrate?

(a) 8.54%,

(b) 17.1%,

(c) 13.7%,

(d) 24.4%,

(e) 82.9%

2. Calculate the percentage of potassium, by mass, in K_2PtCl_6 .

3-4 Avogadro's Number and the Mole



Amedeo Avogadro
(1776-1856)

Avogadro's law:

“Equal volumes of gases under the same conditions of temperature and pressure will contain equal numbers of molecules”



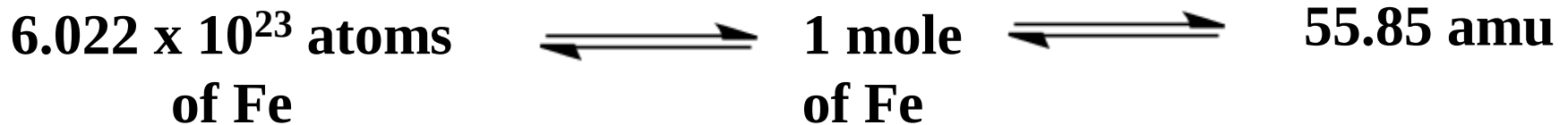


A teaspoon of water contains about 2×10^{23} water molecules
a so large number!

➔ Need a chemical unit for the amount of a substance

Mole

- number of atoms in exactly 12 g of isotopically pure ^{12}C
- **Avogadro's number**, N_A , $6.02214129 \times 10^{23}$
 - We use **atoms** when dealing with a single, individual element.
 - We use **molecules** or **particles** when dealing with more than one element or a compound



EXAMPLE

Without using a calculator, arrange these samples in order of increasing numbers of carbon atoms: 12 g ^{12}C , 1 mol C_2H_2 , 9×10^{23} molecules of CO_2

- 12 g of ^{12}C contains 1 mol of C atoms = 6.02×10^{23} C atoms
- 1 mol of C_2H_2 contains 6.02×10^{23} C_2H_2 molecules, $2 \times 6.02 \times 10^{23}$ C atoms
- 9×10^{23} molecules of CO_2 contains 9×10^{23} C atoms

The order is 12 g of ^{12}C (6.02×10^{23} C atoms) < 9×10^{23} molecules of CO_2 (9×10^{23} C atoms) < 1 mol of C_2H_2 (12.04×10^{23} C atoms)

PRACTICE

1. Determine which of the following samples contains the fewest sodium atoms?

- (a) 1 mol sodium oxide,
- (b) 45 g sodium fluoride,
- (c) 50 g sodium chloride,
- (d) 1 mol sodium nitrate?

2. Without using a calculator, arrange these samples in order of increasing numbers of O atoms: 1 mol H_2O , 1 mol CO_2 , 3×10^{23} molecules of O_3 .

EXAMPLE

Calculate the number of H atoms in 0.350 mol of $\text{C}_6\text{H}_{12}\text{O}_6$.

Moles $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow$ # molecules $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow$ # atoms H

$$\begin{aligned}\text{H atoms} &= (0.350 \text{ mol } \text{C}_6\text{H}_{12}\text{O}_6) \left(\frac{6.02 \times 10^{23} \text{ molecules } \text{C}_6\text{H}_{12}\text{O}_6}{1 \text{ mol } \text{C}_6\text{H}_{12}\text{O}_6} \right) \left(\frac{12 \text{ H atoms}}{1 \text{ molecule } \text{C}_6\text{H}_{12}\text{O}_6} \right) \\ &= 2.53 \times 10^{24} \text{ H atoms}\end{aligned}$$

PRACTICE

1. How many sulfur atoms are in

(a) 0.45 mol BaSO_4

(b) 1.10 mol of aluminum sulfide

2. How many oxygen atoms are in

(a) 0.25 mol $\text{Ca}(\text{NO}_3)_2$

(b) 1.50 mol of sodium carbonate

Molar Mass

^{12}C has an atomic weight of 12 amu

1 mole of ^{12}C has a mass of 12 g

The **atomic weight** of an element in atomic mass units is **numerically equal to** the mass in grams of 1 mol of that element

How about 1 mole of Cl and 1 mole of NaCl?

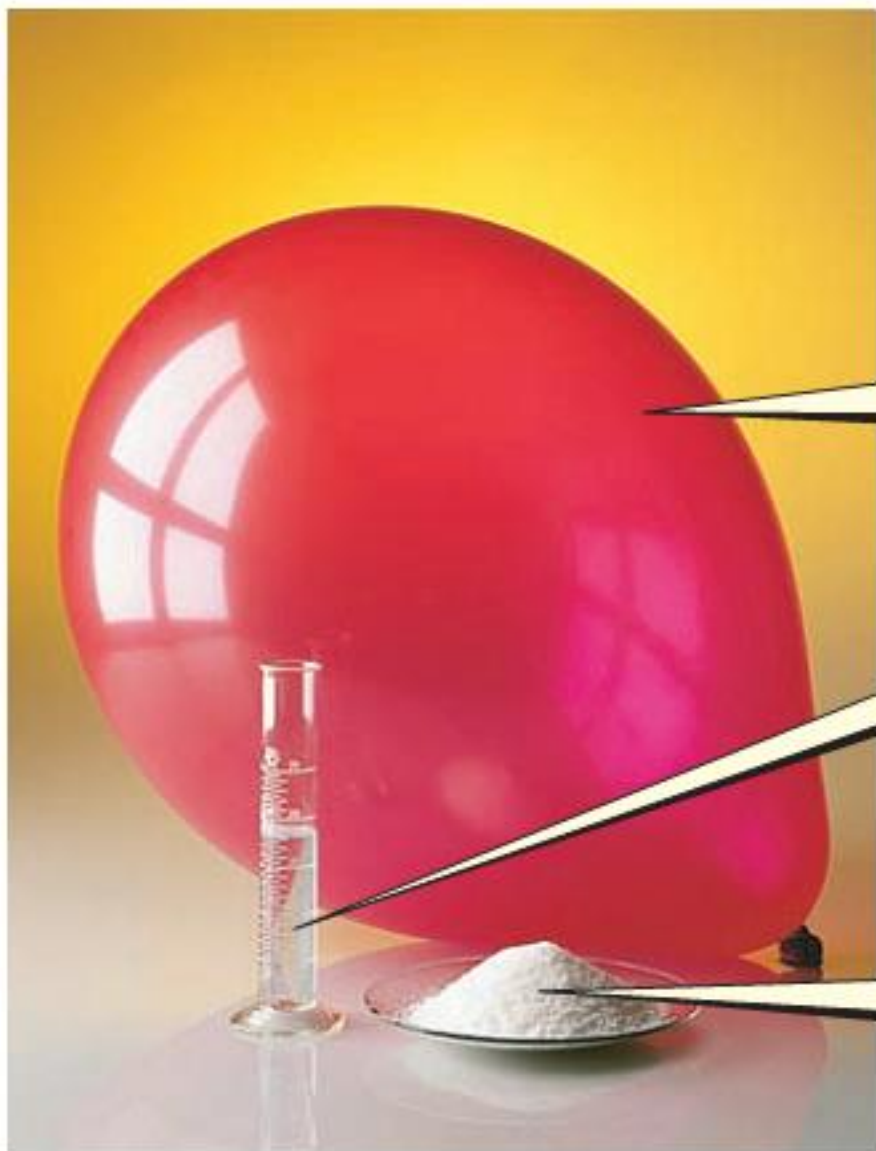
Cl has an atomic weight of 35.5 amu → mol Cl has a mass of 35.5 g.

Au has an atomic weight of 197 amu → mol Au has a mass of 197 g.

H_2O has a formula weight of 18.0 amu → mol H_2O has a mass of 18.0 g

NaCl has a formula weight of 58.5 amu → mol NaCl has a mass of 58.5 g

Name of Substance	Formula	Formula Weight (amu)	Molar Mass (g/mol)	Number and Kind of Particles in One Mole
Atomic nitrogen	N	14.0	14.0	6.02×10^{23} N atoms
Molecular nitrogen	N ₂	28.0	28.0	$\left\{ \begin{array}{l} 6.02 \times 10^{23} \text{ N}_2 \text{ molecules} \\ 2(6.02 \times 10^{23}) \text{ N atoms} \end{array} \right.$
Silver	Ag	107.9	107.9	6.02×10^{23} Ag atoms
Silver ions	Ag ⁺	107.9 ^a	107.9	6.02×10^{23} Ag ⁺ ions
Barium chloride	BaCl ₂	208.2	208.2	$\left\{ \begin{array}{l} 6.02 \times 10^{23} \text{ BaCl}_2 \text{ formula units} \\ 6.02 \times 10^{23} \text{ Ba}^{2+} \text{ ions} \\ 2(6.02 \times 10^{23}) \text{ Cl}^- \text{ ions} \end{array} \right.$



1 mol $\text{O}_2(\text{g})$ has a mass of 32.0 g

1 mol $\text{H}_2\text{O}(\text{l})$ has a mass of 18.0 g

1 mol $\text{NaCl}(\text{s})$ has a mass of 58.45 g

EXAMPLE

What is the molar mass of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$.

$$\begin{array}{rcl} 6 \text{ C atoms} & = & 6(12.0 \text{ amu}) = 72.0 \text{ amu} \\ 12 \text{ H atoms} & = & 12(1.0 \text{ amu}) = 12.0 \text{ amu} \\ 6 \text{ O atoms} & = & 6(16.0 \text{ amu}) = 96.0 \text{ amu} \\ & & \hline & & 180.0 \text{ amu} \end{array}$$

$\text{C}_6\text{H}_{12}\text{O}_6$ has a formula weight of 180.0 amu, 1 mol of this substance (6.02×10^{23} molecules) has a mass of 180.0 g.

➔ Glucose has a molar mass of 180.0 g/mol

PRACTICE

1. A sample of an ionic compound containing iron and chlorine is analyzed and found to have a molar mass of 126.8 g/mol. What is the charge of the iron in this compound?

(a) 1+

(b) 2+

(c) 3+

(d) 4+

2. Calculate the molar mass of $\text{Ca}(\text{NO}_3)_2$.

Converting Mass to Mole

Calculate the number of moles of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) in 5.380 g of $\text{C}_6\text{H}_{12}\text{O}_6$

4 significant figures

$$1 \text{ mol } \text{C}_6\text{H}_{12}\text{O}_6 = 180.0 \text{ g } \text{C}_6\text{H}_{12}\text{O}_6$$

$$\begin{aligned} \text{Moles } \text{C}_6\text{H}_{12}\text{O}_6 &= (5.380 \text{ g } \cancel{\text{C}_6\text{H}_{12}\text{O}_6}) \left(\frac{1 \text{ mol } \text{C}_6\text{H}_{12}\text{O}_6}{180.0 \text{ g } \cancel{\text{C}_6\text{H}_{12}\text{O}_6}} \right) \\ &= 0.02989 \text{ mol } \text{C}_6\text{H}_{12}\text{O}_6 \end{aligned}$$

4 significant figures

PRACTICE

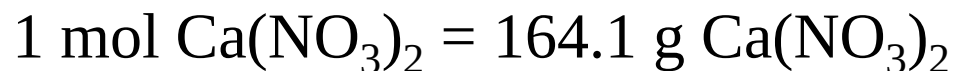
1. How many moles of sodium bicarbonate (NaHCO_3) are in 508 g of NaHCO_3 ?

2. How many moles of water are in 1.00 L of water, whose density is 1.00 g/mL?

Converting Mole to Mass

Calculate the mass, in grams, of 0.433 mol of calcium nitrate

3 significant figures



$$\begin{aligned} \text{Grams Ca(NO}_3)_2 &= (0.433 \text{ mol Ca(NO}_3)_2) \left(\frac{164.1 \text{ g Ca(NO}_3)_2}{1 \text{ mol Ca(NO}_3)_2} \right) \\ &= 71.1 \text{ g Ca(NO}_3)_2 \end{aligned}$$

3 significant figures

PRACTICE

1. What is the mass, in grams, of

(a) 6.33 mol of NaHCO_3

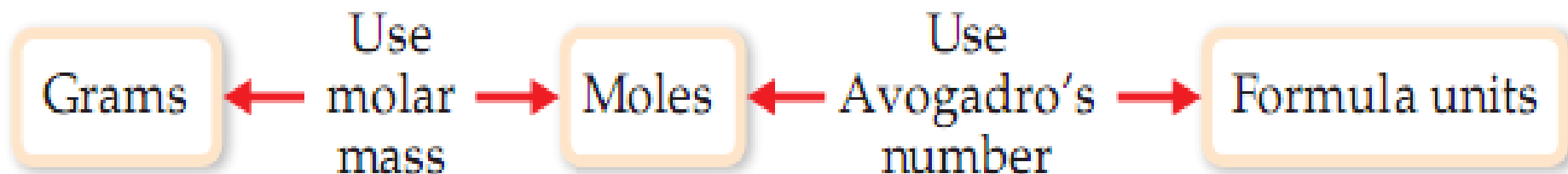
(b) 3.0×10^{-5} mol of sulfuric acid?

2. What is the mass, in grams, of

(a) 0.50 mol of diamond (C)

(b) 0.155 mol of ammonium chloride?

Interconverting Masses and Numbers of Particles



Example: Calculate the number of copper atoms in 3 g of copper

$$\begin{aligned}\text{Cu atoms} &= (3 \text{ g Cu}) \left(\frac{1 \text{ mol Cu}}{63.5 \text{ g Cu}} \right) \left(\frac{6.02 \times 10^{23} \text{ Cu atoms}}{1 \text{ mol Cu}} \right) \\ &= 3 \times 10^{22} \text{ Cu atoms}\end{aligned}$$

PRACTICE

1. How many chlorine atoms are in 12.2 g of CCl_4 ?

(a) 4.77×10^{22}

(b) 7.34×10^{24}

(c) 1.91×10^{23}

(d) 2.07×10^{23}

2. For 4.20 g of HNO_3 ,

(a) How many nitric acid molecules are in this sample?

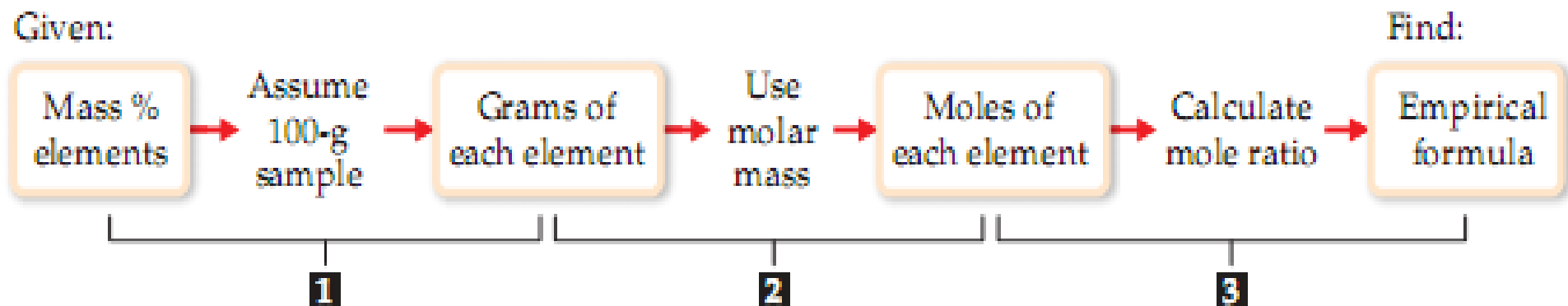
(b) How many O atoms are in this sample?

3-5 Empirical Formulas from Analyses

The **empirical formula** for a substance tells us the **relative number of atoms** of each element in the substance.

*H₂O shows that water contains two H atoms for each O atom
1 mol of H₂O contains 2 mol of H atoms and 1 mol of O atoms*

The **ratio of the numbers of moles** of all elements in a compound gives the **subscripts** in the compound's empirical formula



EXAMPLE

Ascorbic acid (vitamin C) contains 40.92% C, 4.58% H, and 54.50% O by mass. What is the empirical formula of ascorbic acid?



- (1) For simplicity, assume that we have exactly 100 g of material
- (2) Calculate the number of moles of each element

$$\text{Moles C} = (40.92 \text{ g C}) \left(\frac{1 \text{ mol C}}{12.01 \text{ g C}} \right) = 3.407 \text{ mol C}$$

$$\text{Moles H} = (4.58 \text{ g H}) \left(\frac{1 \text{ mol H}}{1.008 \text{ g H}} \right) = 4.54 \text{ mol H}$$

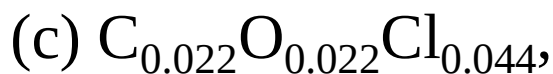
$$\text{Moles O} = (54.50 \text{ g O}) \left(\frac{1 \text{ mol O}}{16.00 \text{ g O}} \right) = 3.406 \text{ mol O}$$

- (3) Determine the simplest whole-number ratio of moles by dividing each number of moles by the smallest number of moles.

$$\text{C: } \frac{3.407}{3.406} = 1.000 \quad \text{H: } \frac{4.54}{3.406} = 1.33 \quad \text{O: } \frac{3.406}{3.406} = 1.000 \quad \text{C: H: O} = (3 \times 1: 3 \times 1.33: 3 \times 1) = (3: 4: 3)$$

PRACTICE

1. A 2.144-g sample of phosgene, a compound used as a chemical warfare agent during World War I, contains 0.260 g of carbon, 0.347 g of oxygen, and 1.537 g of chlorine. What is the empirical formula of this substance?



PRACTICE

2. A 5.325-g sample of methyl benzoate, a compound used in the manufacture of perfumes, contains 3.758 g of carbon, 0.316 g of hydrogen, and 1.251 g of oxygen. What is the empirical formula of this substance?

Molecular Formulas from Empirical Formulas

The **subscripts in the molecular formula** of a substance are always **whole-number multiples of the subscripts** in its empirical formula

$$\text{Whole-number multiple} = \frac{\text{molecular weight}}{\text{empirical formula weight}}$$

Example:

Empirical formula of ascorbic acid is **C₃H₄O₃**

Empirical formula weight of ascorbic acid is **88.0 amu**

Molecular weight of ascorbic acid is **176 amu**

$$\text{Whole-number multiple} = \frac{\text{molecular weight}}{\text{empirical formula weight}} = \frac{176 \text{ amu}}{88.0 \text{ amu}} = 2$$

The molecular formula of ascorbic acid is **C₆H₈O₆**.

EXAMPLE

Mesitylene, a hydrocarbon found in crude oil, has an empirical formula of C_3H_4 and an experimentally determined molecular weight of 121 amu. What is its molecular formula?

(1) Calculate the formula weight of the empirical formula C_3H_4

$$3(12.0 \text{ amu}) + 4(1.0 \text{ amu}) = 40.0 \text{ amu}$$

(2) Calculate the whole number multiple

$$\text{Whole-number multiple} = \frac{\text{molecular weight}}{\text{empirical formula weight}} = \frac{121}{40.0} = 3.03$$

The molecular formula of mesitylene is C_9H_{12}

PRACTICE

1. Cyclohexane, a commonly used organic solvent, is 85.6% C and 14.4% H by mass with a molar mass of 84.2 g/mol. What is its molecular formula?
- (a) C_6H ,
 - (b) CH_2 ,
 - (c) C_5H_{24} ,
 - (d) C_6H_{12} ,
 - (e) C_4H_8 .

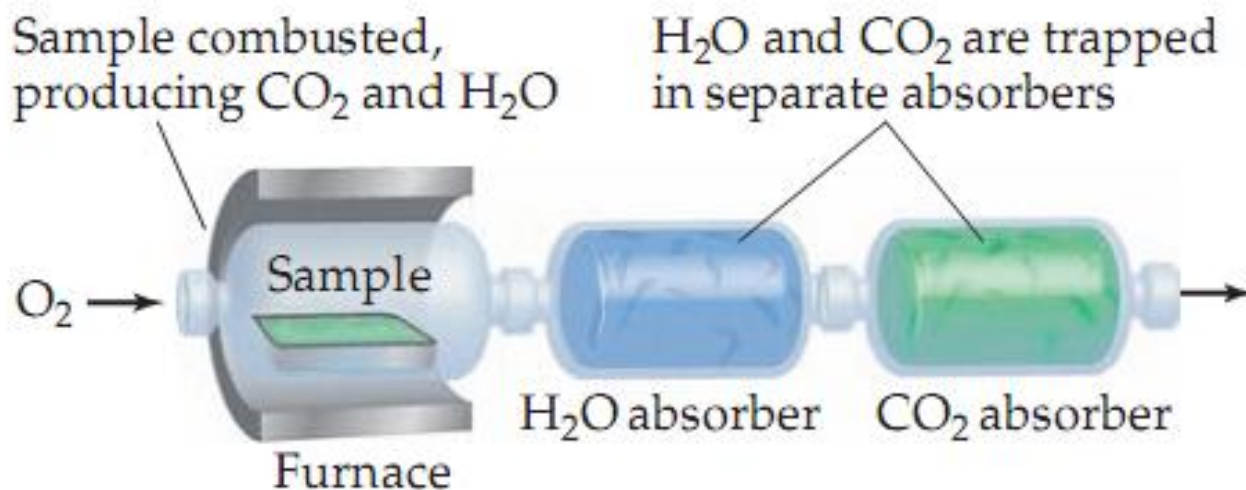
PRACTICE

2. Ethylene glycol, used in automobile antifreeze, is 38.7% C, 9.7% H, and 51.6% O by mass. Its molar mass is 62.1 g/mol.

(a) What is the empirical formula of ethylene glycol?

(b) What is its molecular formula?

Combustion Analysis



Mass gained by each absorber corresponds to mass of CO_2 or H_2O produced

Mass CO_2 produced $\rightarrow$ Molar mass CO_2 44.0 g/mol $\rightarrow$ Moles CO_2 produced $\rightarrow$ 1 C atom per CO_2 molecule $\rightarrow$ Moles of C in original sample $\rightarrow$ Molar mass C 12.0 g/mol $\rightarrow$ Mass C in original sample

Mass H_2O produced $\rightarrow$ Molar mass H_2O 18.0 g/mol $\rightarrow$ Moles H_2O produced $\rightarrow$ 2 H atoms per H_2O molecule $\rightarrow$ Moles H in original sample $\rightarrow$ Molar mass H 1.01 g/mol $\rightarrow$ Mass H in original sample

EXAMPLE

Isopropyl alcohol, sold as rubbing alcohol, is composed of C, H, and O. Combustion of 0.255 g of isopropyl alcohol produces 0.561 g of CO_2 and 0.306 g of H_2O . Determine the empirical formula of isopropyl alcohol.

(1) Calculate mass of carbon in the original sample

$$\begin{aligned}\text{Grams C} &= (0.561 \text{ g } \cancel{\text{CO}_2}) \left(\frac{1 \cancel{\text{ mol CO}_2}}{44.0 \text{ g } \cancel{\text{CO}_2}} \right) \left(\frac{1 \cancel{\text{ mol C}}}{1 \cancel{\text{ mol CO}_2}} \right) \left(\frac{12.0 \text{ g C}}{1 \cancel{\text{ mol C}}} \right) \\ &= 0.153 \text{ g C}\end{aligned}$$

(2) Calculate mass of hydrogen in the original sample

$$\begin{aligned}\text{Grams H} &= (0.306 \text{ g } \cancel{\text{H}_2\text{O}}) \left(\frac{1 \cancel{\text{ mol H}_2\text{O}}}{18.0 \text{ g } \cancel{\text{H}_2\text{O}}} \right) \left(\frac{2 \cancel{\text{ mol H}}}{1 \cancel{\text{ mol H}_2\text{O}}} \right) \left(\frac{1.01 \text{ g H}}{1 \cancel{\text{ mol H}}} \right) \\ &= 0.0343 \text{ g H}\end{aligned}$$

(3) Calculate mass of oxygen in the original sample

$$\begin{aligned}\text{Mass of O} &= \text{mass of sample} - (\text{mass of C} + \text{mass of H}) \\ &= 0.255 \text{ g} - (0.153 \text{ g} + 0.0343 \text{ g}) = 0.068 \text{ g O}\end{aligned}$$

EXAMPLE

(4) Calculate the number of moles of C, H, and O in the sample

$$\text{Moles C} = (0.153 \text{ g}\cancel{\text{C}}) \left(\frac{1 \text{ mol C}}{12.0 \text{ g}\cancel{\text{C}}} \right) = 0.0128 \text{ mol C}$$

$$\text{Moles H} = (0.0343 \text{ g}\cancel{\text{H}}) \left(\frac{1 \text{ mol H}}{1.01 \text{ g}\cancel{\text{H}}} \right) = 0.0340 \text{ mol H}$$

$$\text{Moles O} = (0.068 \text{ g}\cancel{\text{O}}) \left(\frac{1 \text{ mol O}}{16.0 \text{ g}\cancel{\text{O}}} \right) = 0.0043 \text{ mol O}$$

(5) Find the empirical formula

$$\text{C:} \frac{0.0128}{0.0043} = 3.0 \quad \text{H:} \frac{0.0340}{0.0043} = 7.9 \quad \text{O:} \frac{0.0043}{0.0043} = 1.0$$

The first two numbers are very close to the whole numbers 3 and 8, giving the empirical formula **C₃H₈O**.

PRACTICE

1. The compound dioxane, which is used as a solvent in various industrial processes, is composed of C, H, and O atoms. Combustion of a 2.203-g sample of this compound produces 4.401 g CO_2 and 1.802 g H_2O . A separate experiment shows that it has a molar mass of 88.1 g/mol. Which of the following is the correct molecular formula for dioxane?
- (a) $\text{C}_2\text{H}_4\text{O}$,
 - (b) $\text{C}_4\text{H}_4\text{O}_2$,
 - (c) CH_2 ,
 - (d) $\text{C}_4\text{H}_8\text{O}_2$.

PRACTICE

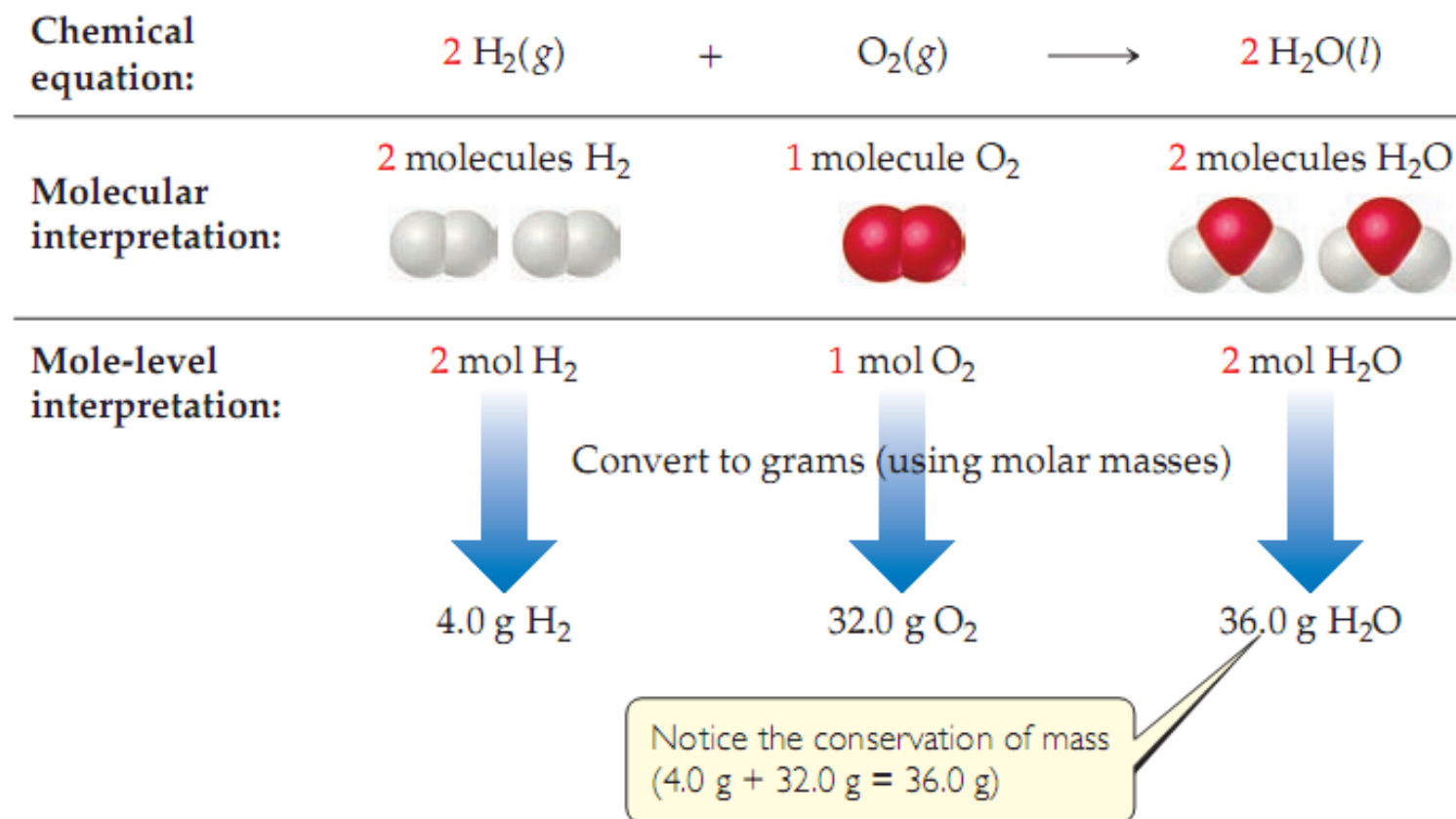
2. Caproic acid, responsible for the odor of dirty socks, is composed of C, H, and O atoms. Combustion of a 0.225-g sample of this compound produces 0.512 g CO_2 and 0.209 g H_2O . Caproic acid has a molar mass of 116 g/mol.

(a) What is the empirical formula of caproic acid?

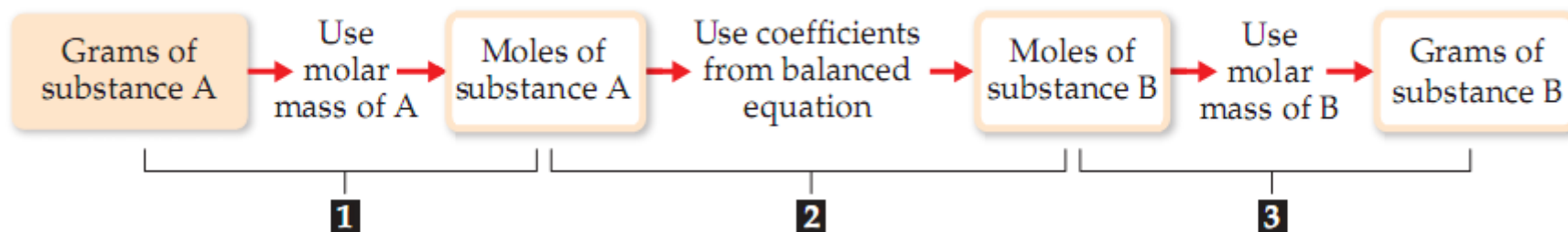
(b) What is the molecular formula of caproic acid?

3-6 Quantitative Information from Balanced Equations

The **coefficients** in a balanced chemical equation indicate both the **relative numbers of molecules** (or formula units) in the reaction and the **relative numbers of moles**



Given:



Find:

Consider the combustion of 1.00 g of butane (C_4H_{10}), calculate the mass of CO_2 produced



$$\text{Moles C}_4\text{H}_{10} = (1.00 \text{ g C}_4\text{H}_{10}) \left(\frac{1 \text{ mol C}_4\text{H}_{10}}{58.0 \text{ g C}_4\text{H}_{10}} \right) = 1.72 \times 10^{-2} \text{ mol C}_4\text{H}_{10}$$

$$\text{Moles CO}_2 = (1.72 \times 10^{-2} \text{ mol C}_4\text{H}_{10}) \left(\frac{8 \text{ mol CO}_2}{2 \text{ mol C}_4\text{H}_{10}} \right) = 6.88 \times 10^{-2} \text{ mol CO}_2$$

$$\text{Grams CO}_2 = (6.88 \times 10^{-2} \text{ mol CO}_2) \left(\frac{44.0 \text{ g CO}_2}{1 \text{ mol CO}_2} \right) = 3.03 \text{ g CO}_2$$

PRACTICE

1. Sodium hydroxide reacts with carbon dioxide to form sodium carbonate and water. How many grams of Na_2CO_3 can be prepared from 2.40 g of NaOH?
- (a) 3.18 g,
 - (b) 6.36 g,
 - (c) 1.20 g,
 - (d) 0.0300 g

PRACTICE

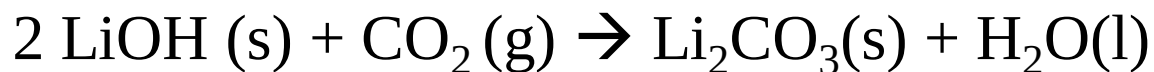
Decomposition of KClO_3 is sometimes used to prepare small amounts of O_2 in the laboratory



How many grams of O_2 can be prepared from 4.50 g of KClO_3 ?

Calculating amounts of Reactants and products

Solid lithium hydroxide is used in space vehicles to remove the carbon dioxide gas exhaled by astronauts. The hydroxide reacts with the carbon dioxide to form solid lithium carbonate and liquid water. How many grams of carbon dioxide can be absorbed by 1.00 g of lithium hydroxide?



molar mass of LiOH: $16.94 + 16.00 + 1.01 = 23.95 \text{ g/mol}$

molar mass of CO_2 : $12.01 + 2(16.00) = 44.01 \text{ g/mol}$

$$(1.00 \text{ g LiOH}) \left(\frac{1 \text{ mol LiOH}}{23.95 \text{ g LiOH}} \right) \left(\frac{1 \text{ mol CO}_2}{2 \text{ mol LiOH}} \right) \left(\frac{44.01 \text{ g CO}_2}{1 \text{ mol CO}_2} \right) = 0.919 \text{ g CO}_2$$

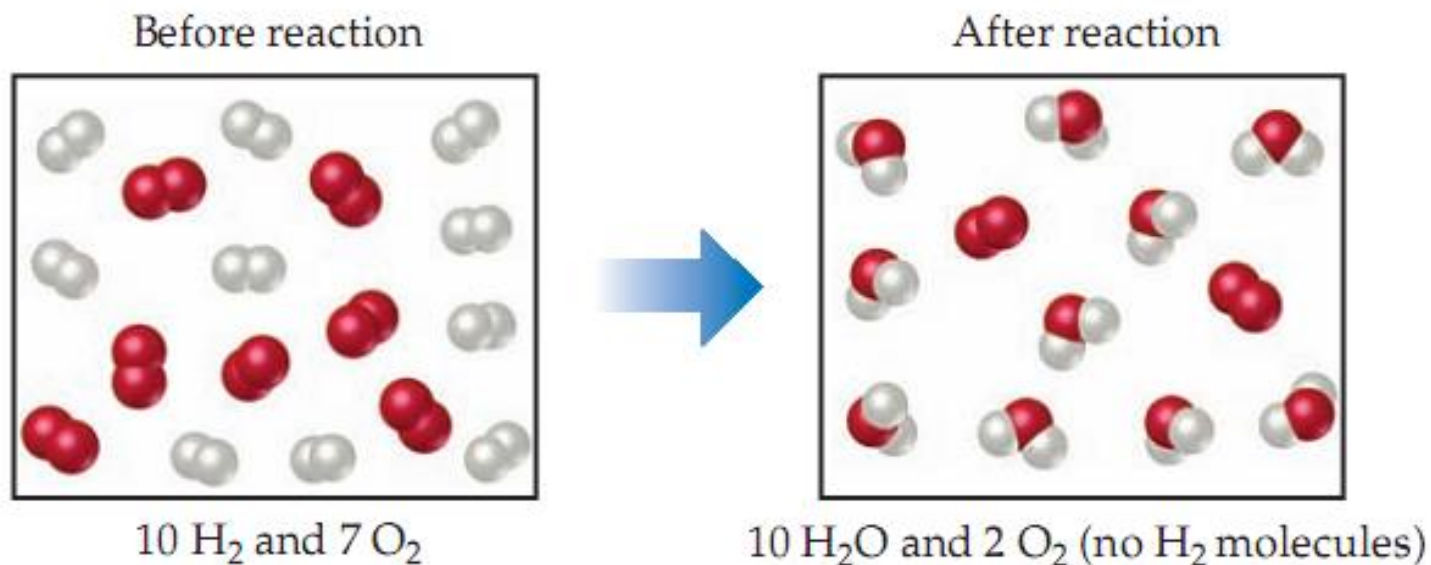
PRACTICE

1. Propane, C_3H_8 , is a common fuel used for cooking and home heating. What mass of O_2 is consumed in the combustion of 1.00 g of propane?
 - (a) 5.00 g,
 - (b) 0.726 g,
 - (c) 2.18 g,
 - (d) 3.63 g.

PRACTICE

2. Methanol, CH_3OH , reacts with oxygen from air in a combustion reaction to form water and carbon dioxide. What mass of water is produced in the combustion of 23.6 g of methanol?

3-6 Limiting Reactants



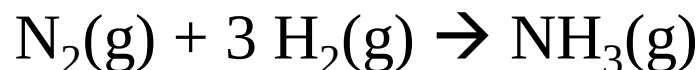
Because H₂ is completely consumed, it is the limiting reactant.

Because some O₂ is left over after the reaction is complete, it is the excess reactant.

➔ The amount of H₂O formed depends on the amount of limiting reactant, H₂

EXAMPLE

The most important commercial process for converting N_2 from the air into nitrogen-containing compounds is based on the reaction of N_2 and H_2 to form ammonia (NH_3). How many moles of NH_3 can be formed from 3.0 mol of N_2 and 6.0 mol of H_2 ?



- (1) The number of moles of H_2 needed for complete consumption of 3.0 mol of N_2 is

$$\text{Moles H}_2 = (3.0 \text{ mol } \cancel{\text{N}_2}) \left(\frac{3 \text{ mol H}_2}{1 \text{ mol } \cancel{\text{N}_2}} \right) = 9.0 \text{ mol H}_2$$

- (2) The quantity of NH_3 produced

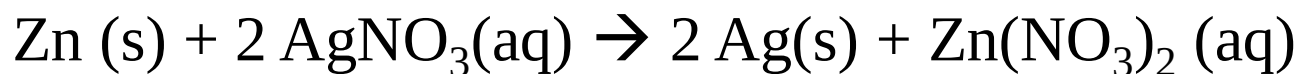
$$\text{Moles NH}_3 = (6.0 \text{ mol } \cancel{\text{H}_2}) \left(\frac{2 \text{ mol NH}_3}{3 \text{ mol } \cancel{\text{H}_2}} \right) = 4.0 \text{ mol NH}_3$$

PRACTICE

1. When 24 mol of methanol and 15 mol of oxygen combine in the combustion reaction $2 \text{CH}_3\text{OH}(\text{l}) + 3 \text{O}_2(\text{g}) \rightarrow 2 \text{CO}_2(\text{g}) + 4 \text{H}_2\text{O}(\text{g})$, what is the excess reactant and how many moles of it remains at the end of the reaction?
- (a) 9 mol $\text{CH}_3\text{OH}(\text{l})$,
 - (b) 10 mol $\text{CO}_2(\text{g})$,
 - (c) 10 mol $\text{CH}_3\text{OH}(\text{l})$,
 - (d) 14 mol $\text{CH}_3\text{OH}(\text{l})$,
 - (e) 1 mol $\text{O}_2(\text{g})$.

PRACTICE

2. When a 2.00-g strip of zinc metal is placed in an aqueous solution containing 2.50 g of silver nitrate, the reaction is



- (a) Which reactant is limiting?
- (b) How many grams of Ag form?
- (c) How many grams of $\text{Zn(NO}_3)_2$ form?
- (d) How many grams of the excess reactant are left at the end of the reaction?

Theoretical and Percent Yields

$$\text{Percent yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

Adipic acid, $\text{H}_2\text{C}_6\text{H}_8\text{O}_4$, used to produce nylon, is made commercially by a reaction between cyclohexane (C_6H_{12}) and O_2 :



- (a) Assume that you carry out this reaction with 25.0 g of cyclohexane and that cyclohexane is the limiting reactant. What is the theoretical yield of adipic acid?
- (b) If you obtain 33.5 g of adipic acid, what is the percent yield for the reaction?

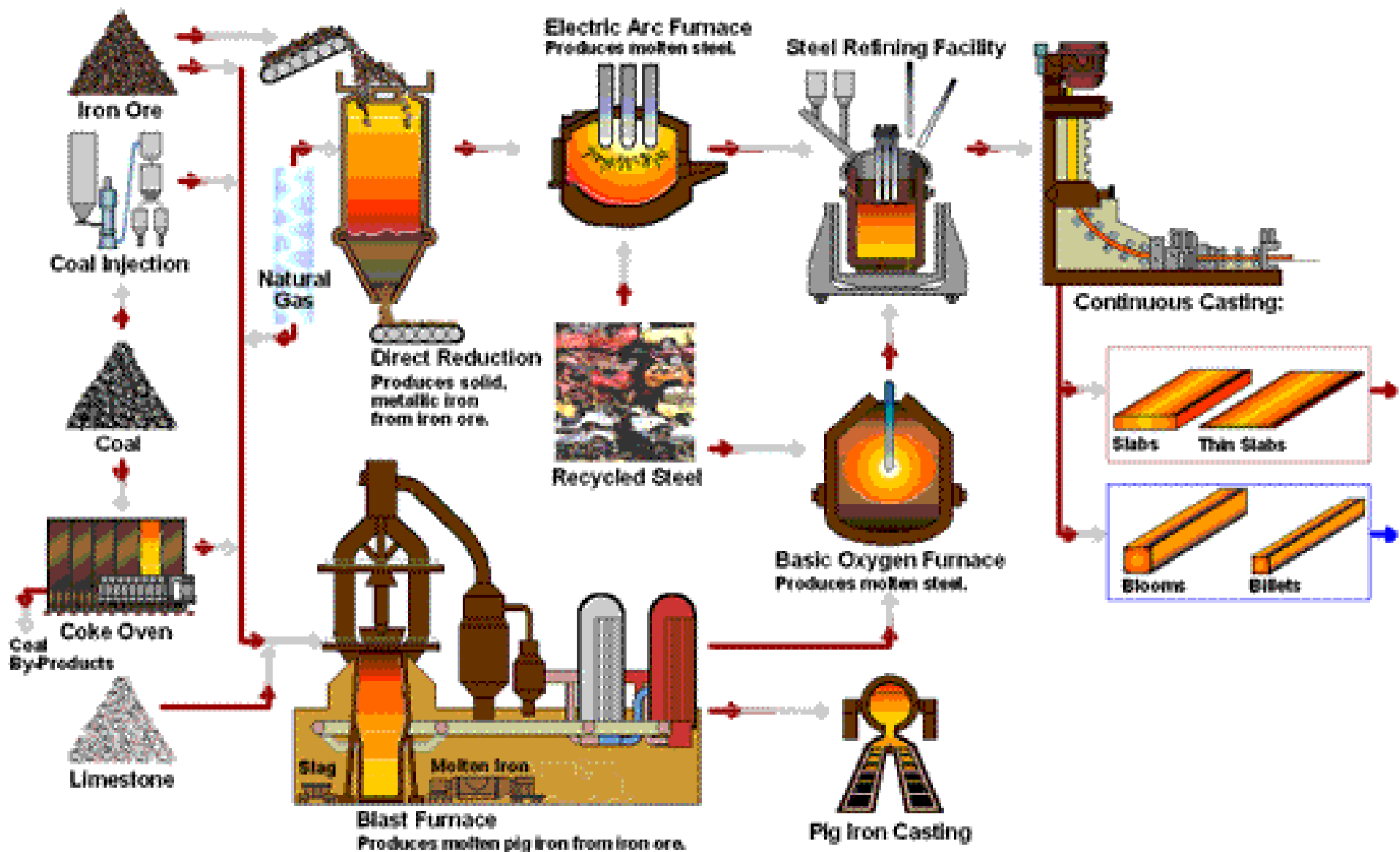
(a) The theoretical yield is

$$\begin{aligned} \text{Grams H}_2\text{C}_6\text{H}_8\text{O}_4 &= (25.0 \text{ g } \cancel{\text{C}_6\text{H}_{12}}) \left(\frac{1 \cancel{\text{mol C}_6\text{H}_{12}}}{84.0 \text{ g } \cancel{\text{C}_6\text{H}_{12}}} \right) \left(\frac{2 \cancel{\text{mol H}_2\text{C}_6\text{H}_8\text{O}_4}}{2 \cancel{\text{mol C}_6\text{H}_{12}}} \right) \left(\frac{146.0 \text{ g H}_2\text{C}_6\text{H}_8\text{O}_4}{1 \cancel{\text{mol H}_2\text{C}_6\text{H}_8\text{O}_4}} \right) \\ &= 43.5 \text{ g H}_2\text{C}_6\text{H}_8\text{O}_4 \end{aligned}$$

$$\text{(b) Percent yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\% = \frac{33.5 \text{ g}}{43.5 \text{ g}} \times 100\% = 77.0\%$$

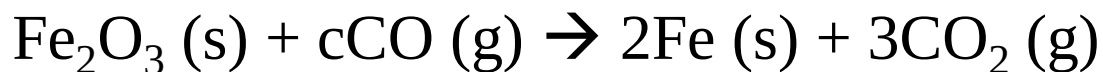
PRACTICE

1. If 3.00 g of titanium metal is reacted with 6.00 g of chlorine gas, Cl_2 , to form 7.7 g of titanium (IV) chloride in a combination reaction, what is the percent yield of the product?
 - (a) 65%,
 - (b) 96%,
 - (c) 48%,
 - (d) 86%.



PRACTICE

2. Imagine you are working on ways to improve the process by which iron ore containing Fe_2O_3 is converted into iron:



(a) If you start with 150 g of Fe_2O_3 as the limiting reactant, what is the theoretical yield of Fe?

(b) If your actual yield is 87.9 g, what is the percent yield?

Homeworks

Exercises:

3.5

3.13

3.21

3.23

3.37

3.51

3.63

3.81

3.91

Chemistry

The Central Science

Fourteenth Edition in SI Units

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You will have a quiz next week

What are allowed?

- A pen
- A periodic table
- An A4-sized sheet of hand written notes
- A calculator

No other devices are allowed!